



UTM

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SECV2223-Web Programming

Assignment 1

Comparative Evaluation of News Websites Using Google Lighthouse

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Abstract

In the digital era, news websites have a critical part to ensure that information is delivered efficiently to their audience. The performance of websites will directly affect their user experience, accessibility, and ranking within search engines. So, it is important for us to research and analysis websites performance. In this research, we selected and evaluates news website of Malaysia with the help of google Lighthouse. We analyzed those websites with four performance metrics, including Performance, Accessibility, Best Practices, and SEO. Based on results obtained, we observed that these websites differ in terms of performance because of several reasons such as heavy content and lack of technology.

Keywords

Website Performance, Google Lighthouse, SEO, Accessibility, News Websites

1.0 Introduction

Today's digital world has seen considerable change in how people access the news rather than accessing news from traditional media sources, with more reliance on the internet for news purposes. News websites have enabled users to access news at any time and from anywhere because of the nature of technology, which allows for instant updates and a wide variety of sources that cater to diverse interests and preferences.

The performance of websites is significant, and it will affect the satisfaction, involvement, and retention of users. A poorly optimized website will disappoint users and may cause users to leave the website before the page fully loads up. This necessitates the importance of testing and measuring website performance.

Google Lighthouse will be used to measure the performance of selected news websites in Malaysia.

2.0 Methodology

This report uses Google Lighthouse to evaluate the website in four areas: Performance, Accessibility, Best Practices, and SEO. All tests were conducted in Desktop mode and using default settings.

The following core metrics to evaluate each websites :

- Performance: includes First Contentful Paint (FCP), Largest Contentful Paint (LCP), cumulative layout shift, Total Blocking Time and Speed Index
- Accessibility: name and labels, contrast and navigation support
- Best Practice: Accesses site security, use of HTTPS, deprecated APIs, and efficient asset loading.
- SEO: Checks for meta description, crawlability, link structure, and mobile-friendliness indicators.

3.0 Selected Websites

We selected six news websites to analyze; these portals were selected because they were identified as the top six most visited online news websites in Malaysia, according to the Reuters Institute Digital News Report's article, "Malaysia: Digital News Report 2024". The following websites selected are Malaysiakini, Astro Awani, Berita Harian, Harian Metro, The Star, and Sinar Harian.

4.0 Results and Discussion

4.1 Google Lighthouse Results

Website	Performance	Accessibility	Best Practices	SEO
Malaysiakini	72	83	77	77
Astro Awani	47	86	77	77
Berita Harian	57	79	77	83
Harian Metro	20	79	77	92
The Star	46	84	73	N/A
Sinar Harian	58	82	73	92

Table 1: Result of each website results using Google Lighthouse Tool

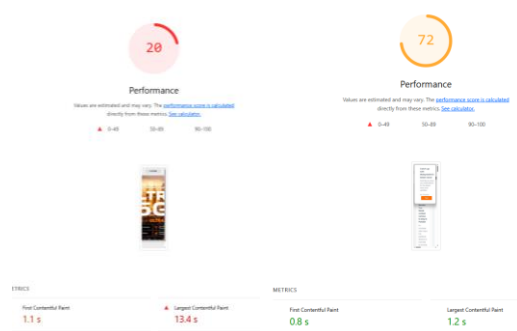
All the summary report of each website are stored in this link:

<https://drive.google.com/drive/folders/1x6W-9F47Dlvbl5p36gach6R5lYqCjA0u>

4.2 Performance Analysis

Based on the results obtained from the Lighthouse tool, all website shows similar performance ,which ranging from among all the websites, we found that the best performance website is Malaysiakini, which achieving the score of 72 .This due to it could maintain a very fast LCP of 1.2seconds, which make it has higher score than others.

While the Harian Metro has recorded the lowest performance score of 20 due to its extremely slow LCP, which is 13.4 seconds and high CLS of 0.402, which makes the loading experience slower.



4.3 Accessibility Analysis

For the accessibility of all websites has shown a higher scores, which ranging from 79 to 86. Among all the websites, the website has highest accessibility are Astro Awani which reach the scores of 86 because it had implemented many labels, which improve user easier to navigate the website .While the Berita Harian and Harian Metro both have a lower scores of 79 due to their buttons does not have accessible names.

4.4 Best Practices Analysis

From the analysis of Lighthouse Tool, all websites show average success in best practice measures, with scores varying from 73 to 77. These scores indicate that the sites follow standard web development guidelines. However, The Star and Sinar Harian has achieved a lower score of 73 compared to other websites that have scores of 77 caused by using deprecated APIs and logging browser errors to the console.

4.5 SEO Analysis

Most websites have strong scores for SEO measures. Harian Metro and Sinar Harian have the highest scores for SEO, both scoring 92. It shows the successful use of SEO techniques by both media organizations. However, it should be noted that The Star failed to provide information about SEO measures, possibly due to audit limitations.

4.6 Overall Comparison

The overall analysis based on the table reveals a clear divide in technical quality across these news sites. Malaysiakini is the highest performer overall, due to it is the most balanced platform with a performance score of 72 while staying competitive in Accessibility and Best Practices. In contrast, Harian Metro is the lowest performer overall, struggling with a critical lack of optimization that results in the slowest load times which performance score of 20 and the weakest accessibility scores although its SEO has performed best among the sites which is score of 90.

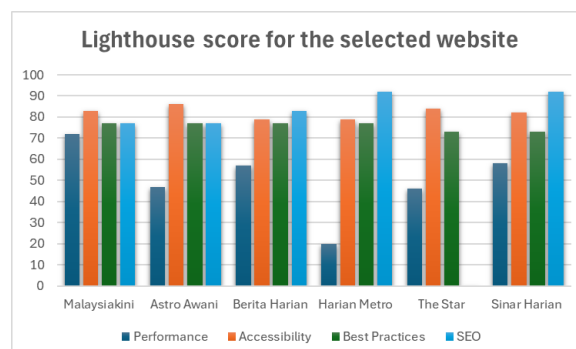


Table 2: Comparison of each website based on their scores

4.7 Recommendations for Optimization

Based on the results of the Lighthouse analysis, several actionable recommendations can be made to improve the overall performance, accessibility, and SEO effectiveness of these websites. Most websites have a weak performance, so it is important to reduce unnecessary JavaScript and optimize image size. For accessibility, it is recommended to add alt descriptions to images and optimize the navigation structure to improve user experience. For best Practices, it should use less “old code” to ensure the website security and a good code structure. For SEO, make the website easier for search engines to find by optimizing content.

5.0 Conclusion

This report uses Google Lighthouse to make a comparative analysis, mainly from performance, accessibility, best practices, and SEO.

From the analysis, different websites have different results. Malaysiakini has a relatively balanced performance, while Harian Metro has the worst performance.

Overall, most websites have good accessibility and SEO, but there is still considerable room for improvement in terms of performance optimization.

6.0 Reference

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